

Bryce Lloyd
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EDUCATION

Johns Hopkins University

B.S. Chemical & Biomolecular Engineering

- GPA: 3.94 | MCAT: 520 | 8-Time Dean's List

Baltimore, Maryland

Graduated May 2026

TECHNICAL SKILLS

- Laboratory: Adherent Cell Culture, Immune Cell Culture, Lentivirus Production, Transduction, 3D Migration Gels, Brightfield Microscopy, Phase Contrast Microscopy, Molecular Cloning, Bacterial Transformation, Transfection, Incucyte Spheroid Cell-Killing Assays, Flow Cytometry, Aseptic Technique
- Computer Programming: Python, Java, HTML, CSS

RESEARCH EXPERIENCE

The Rezvani Lab at MD Anderson Cancer Center

Undergraduate Research Assistant

Houston, Texas

June 2025 – Present

- Leading a project investigating mechanisms by which hypoxic culture enhances CAR-NK cytotoxicity in hostile tumor microenvironments.
- Designed, presented, and conducted experiments evaluating immunometabolic conditioning of CAR-NK cells under nutrient-deficient and hypoxic culture conditions.
- Analyzed cytotoxicity and phenotyping data, presented findings to the Cell Therapy Institute and CPRIT-CURE Summer Program, and contributed to a manuscript in preparation.

The Wirtz Lab at Johns Hopkins University

Undergraduate Research Assistant

Baltimore, Maryland

August 2024 – Present

- Leading a project on developing a novel CAR-T therapy approach from scratch for an autoimmune disease
- Collaborating with graduate student Eban Hanna on his PhD thesis project investigating the use of Velocity Receptors (VR) to enhance CAR-T cell motility in vitro, aiming to improve their ability to penetrate the tumor microenvironment
- Conducting lentivirus production for CAR-T and VR-CAR-T constructs, transducing T-cells, and comparing their migratory behavior in collagen gels, observing that VR-CAR-T cells exhibit elevated migratory phenotypes through Brightfield microscopy imaging with the Nikon Ti2 microscope

AstraZeneca

CAR-T Cell Therapy Intern

Gaithersburg, Maryland

May 2024 – August 2024

- Tasked with writing, testing, and optimizing a procedure for rescuing cell suspension from process equipment used in the CAR-T Cell Therapy clinical manufacturing process in certain failure scenarios
- Identified an unexpected finding during apheresis processing: suggested that in cases of equipment failure, material should be stored at room temperature for up to 5 hours instead of 2-8°C. Presented these findings to the Cell Therapy team and implemented updates to the standard operating procedure (SOP)
- Created an Excel tool to automate cell count calculations for the manufacturing team at the Cell Therapy Facility
- Carried out full scale CAR-T manufacturing processes to support non-clinical material generation requests

PUBLICATIONS, MANUSCRIPTS, PRESENTATIONS

Manuscripts in Preparation

- Hanna E., Lloyd B., Forjaz, A., Min C., Dance Y., Etoh I., Jiang X., Quiergo V., Nair P., Duarte V., Joshi S., Konstantopoulos K., Phillip J., Fan R., Wirtz D. *Synthetic cytokine receptors enhance CAR-T cell functions against TNBC.*
- Lloyd B., Munir M., Moore M., Tiberti S., Daher M., Rezvani K. *Immunometabolic conditioning of NK cells enhances cytotoxicity in hostile tumor microenvironments.*
- Lloyd B., Jiang X., Hanna E., Quiergo V., Wirtz D. *A dual-antigen CAR-T strategy to improve specificity in psoriasis immunotherapy.*

Abstracts and Presentations

- Lloyd B. *Immunometabolic conditioning of NK cells enhances cytotoxicity in hostile tumor microenvironments.* Abstract and poster presentation, CPRIT-CURE Summer Program, MD Anderson Cancer Center, Aug. 2025.
- Lloyd B. Rezvani Lab Research Presentation, MD Anderson Cancer Center, July 2025.
- Lloyd B. *Hold time-based salvage procedure for CAR-T product rescue.* AstraZeneca Poster Presentation, AstraZeneca, Aug. 2024.
- Lloyd B. Cell Therapy Department Presentation, AstraZeneca, Aug. 2024.

CLINICAL EXPERIENCE

University of Maryland Medical Center

UMMBassador Volunteer

Baltimore, Maryland
August 2025 – June 2026

- Rounded on high-need inpatient medicine and surgical units to provide companionship and nonclinical patient support.
- Created “Getting to Know Me” boards highlighting patients’ interests, hometowns, and personal details to help care teams build rapport beyond the diagnosis.

MedStar Health

Student Guest Ambassador

Baltimore, Maryland
August 2025 – June 2026

- Supported inpatient units by orienting patients and families to the hospital environment, explaining call systems, and relaying nonurgent needs to staff.
- Offered flexible bedside companionship, adapting interactions to each patient’s needs, from brief check-ins to longer supportive conversations.

Physician Shadowing

Student Observer

Houston, Texas
June 2023 – Present

- Observed inpatient rounds, outpatient visits, longitudinal follow-up, and translational care, including physician-scientist involvement in cell therapy trials.
- Dr. Joseph Jankovic, MD, Neurology | Dr. May Daher, MD, Hematology/Oncology | Dr. Sachin Kummula, MD, Emergency Medicine | Dr. Dawn Kershner, DO, Cardiology | Dr. Roa Bashtawi, MD, Pediatric Oncology | Dr. Danial Syed, MD, Neurology

TEACHING EXPERIENCE

Johns Hopkins University

Teaching Assistant

Baltimore, Maryland
January 2025 – Present

- Teaching assistant for Introduction to Chemical & Biological Process Analysis (EN.540.202) in the Spring of 2025 and 2026: hold office hours, conduct weekly grading, and proctor exams, 10 hours per week
- Worked with a class of roughly 60 students, supervised by Dr. Nagma Zerín

Johns Hopkins University

Learning Den Personalized Tutor

Baltimore, Maryland
August 2024 – Present

- Tutored 14 students over 4 semesters in Engineering Thermodynamics, Organic Chemistry 1 and 2, and Calculus 1, met weekly to provide supplementary material to aid their studies

EXTRACURRICULAR ACTIVITIES AND LEADERSHIP

- **Johns Hopkins Men's Swim Team (2022-2026):** dedicated 20+ hours per week to practices, lifts, and competitions, competed at the NCAA Division III Championships all 4 years, placed top 8 in the country in the 200 backstroke twice (2025-2026)
- **Special Olympics Maryland Volunteer Swim Coach (2019-2026):** Coached swimmers with intellectual and developmental disabilities, adapting instruction to build water safety, confidence, technique, and joy in competition.
- **Omicron Delta Kappa Leadership Honors Society Event Coordinator (2024-2026):** Organized and ran a service event titled “PB and Jams”, 100 students attended to make 1,100 peanut butter and jelly sandwiches to support a local Baltimore food bank.
- **The Legacy Project Johns Hopkins Chapter (2025-2026):** Documented the life story of an older Turner Station resident for a community book while participating in resident-led efforts to preserve and support the neighborhood.
- **AIChE Big/Little Mentorship Program Mentor (2024-2026):** Mentored 3 underclassmen in chemical engineering through academic guidance, course advice, and career support.
- **Alpha Kappa Psi Business Fraternity (2023-2026):** Participated in professional development, service, and leadership programming with a business-focused student organization.

HONORS AND AWARDS

- CSC Academic All American 2nd Team: One of 30 student-athletes in all of Division III selected
- 4 Time NCAA First Team Scholastic All-American; 8 Time NCAA Division III All-American Honors (2023-2026)
- College Sports Communicators (CSC) All-District Honors (2023-2026)
- Tau Beta Pi Engineering Honors Society | Chi Alpha Sigma | Omicron Delta Kappa Leadership Honors Society
- Chemical and Biomolecular Engineering Departmental Honors
- Maryland Selling Scholarship Homewood (2024-2025) | Maryland Senatorial Scholarship (2022-2026)

HOBBIES AND INTERESTS

- Piano, Chess, Triathlons, Swimming, Hiking/Travel, Science Blogging